Linear Algebra IV

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We cover the following topics in this note.

- Eigenvectors and Diagonalization.
- Characteristic polynomial.
- TBA.

Notation.

$\mathbb{F}$ a field.

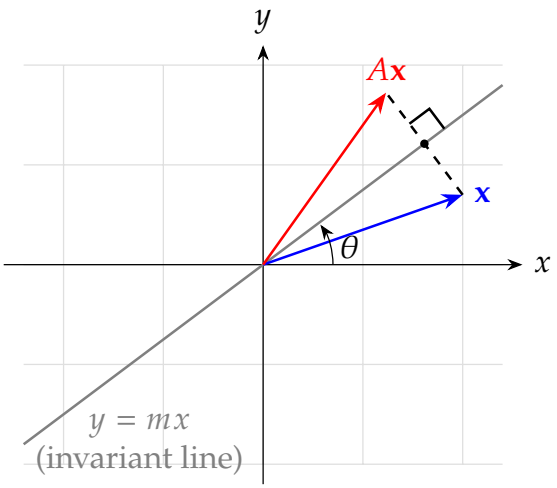
V a finite-dimensional $\mathbb{F}$-vector space.

Observation (Choosing a basis to simplify a linear map). Let $\mathcal{B} = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ be a basis of V such that $[T]_{\mathcal{B}}$ is a diagonal matrix:

$$[T]_{\mathcal{B}} = [T]^{\mathcal{B}}_{\mathcal{B}} = \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{bmatrix}_{n\times n}, \quad \text{i.e.,} \quad T(\mathbf{v}_i) = d_i\mathbf{v}_i \text{ with } 1 \le i \le n.$$

Then T may be very complicated, but with respect to $[T]_{\mathcal{B}}$ it looks nice.

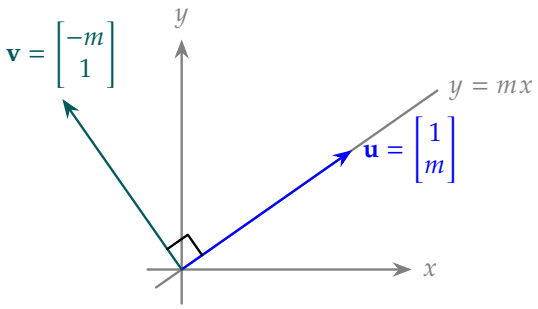
Example 1. Fix $m \in \mathbb{R}$ and let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be the reflection across the line $y = mx$.



Step 1: Choose a basis adapted to the axis. Take

$$\mathbf{u} = \begin{bmatrix} 1 \\ m \end{bmatrix} \quad (\text{along } y = mx), \qquad \mathbf{v} = \begin{bmatrix} -m \\ 1 \end{bmatrix} \quad (\text{perpendicular to } y = mx),$$

since $\mathbf{u} \cdot \mathbf{v} = 1 \cdot (-m) + m \cdot 1 = 0$.



Step 2: Use the defining property of reflection. Reflection across the line

$$L = \text{span}(\mathbf{u}) = \{(t, mt) : t \in \mathbb{R}\}$$

is defined by

- **vectors on the axis** stay fixed, and
- **vectors perpendicular to the axis** reverse direction.

Every vector $\mathbf{w} \in \mathbb{R}^2$ decomposes uniquely as

$$\mathbf{w} = a\,\mathbf{u} + b\,\mathbf{v}.$$

A reflection keeps the parallel component and flips the perpendicular one, so

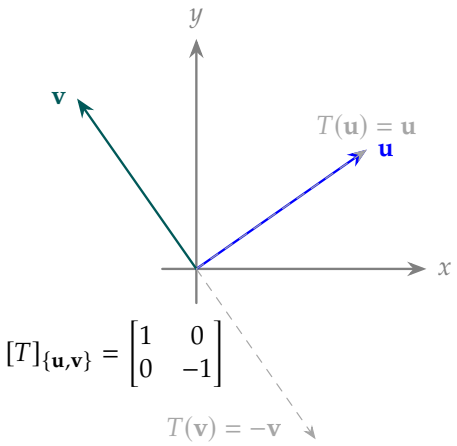
$$T(a\mathbf{u}) = a\mathbf{u} \qquad T(b\mathbf{v}) = -b\mathbf{v}.$$

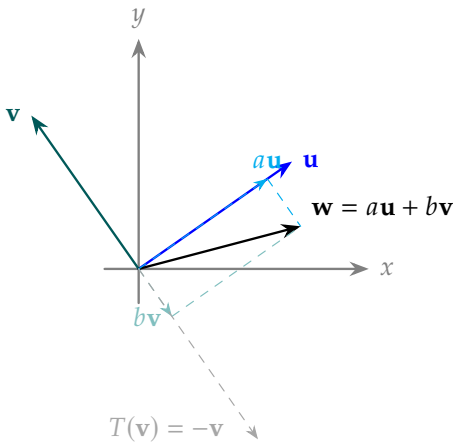
Since T is linear,

$$T(a\mathbf{u} + b\mathbf{v}) = T(a\mathbf{u}) + T(b\mathbf{v}) = a\mathbf{u} - b\mathbf{v}.$$

In the basis $\mathcal{B} = \{\mathbf{u}, \mathbf{v}\}$,

$$[T]_{\mathcal{B}} = \begin{bmatrix} T(\mathbf{u}) & T(\mathbf{v}) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$





Step 3: Convert back to the standard basis. Let $\mathcal{E} = \{\mathbf{e}_1, \mathbf{e}_2\}$ be the standard basis and let $\mathcal{B} = \{\mathbf{u}, \mathbf{v}\}$ be another basis. Define the change-of-coordinates matrix

$$P := [\mathrm{Id}]_{\mathcal{B}}^{\mathcal{E}} = \begin{bmatrix} \mathrm{Id}(\mathbf{u}) & \mathrm{Id}(\mathbf{v}) \end{bmatrix} = \begin{bmatrix} \mathbf{u} & \mathbf{v} \end{bmatrix}.$$

Then for every $\mathbf{x} \in \mathbb{R}^2$ we have

$$[\mathbf{x}]_{\mathcal{E}} = P[\mathbf{x}]_{\mathcal{B}} \qquad \text{and} \qquad [\mathbf{x}]_{\mathcal{B}} = P^{-1}[\mathbf{x}]_{\mathcal{E}}.$$

If $D = [T]_{\mathcal{B}}^{\mathcal{B}}$ is the matrix of T in the basis $\mathcal{B}$, then

$$[T(\mathbf{x})]_{\mathcal{B}} = D[\mathbf{x}]_{\mathcal{B}}.$$

Converting the output back to standard coordinates gives

$$[T(\mathbf{x})]_{\mathcal{E}} = P[T(\mathbf{x})]_{\mathcal{B}} = P\,D\,[\mathbf{x}]_{\mathcal{B}} = P\,D\,P^{-1}[\mathbf{x}]_{\mathcal{E}}.$$

Therefore the standard-basis matrix of T is

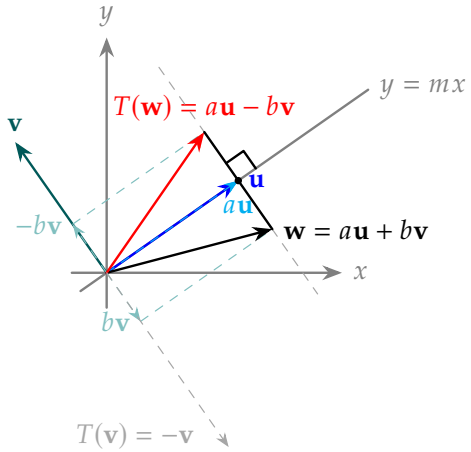
$$\begin{aligned} A := [T]_{\mathcal{E}}^{\mathcal{E}} &= [\mathrm{Id}]_{\mathcal{B}}^{\mathcal{E}}\,[T]_{\mathcal{B}}^{\mathcal{B}}\,[\mathrm{Id}]_{\mathcal{E}}^{\mathcal{B}} \\ &= \underbrace{P}_{\text{Convert back to standard coordinates}} \quad \underbrace{D}_{\text{Apply } T \text{ in } \{\mathbf{u},\mathbf{v}\}\text{-coordinates}} \quad \underbrace{P^{-1}}_{\text{Convert to } \{\mathbf{u},\mathbf{v}\}\text{-coordinates}}. \end{aligned}$$

Compute P^{-1} (since $\det P = 1 + m^2$):

$$P^{-1} = \frac{1}{1+m^2}\begin{bmatrix} 1 & m \\ -m & 1 \end{bmatrix}.$$

Since

$$PD = \begin{bmatrix} 1 & -m \\ m & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & m \\ m & -1 \end{bmatrix},$$



we obtain

$$\begin{aligned} A &= (PD)P^{-1} \\ &= \frac{1}{1+m^2}\begin{bmatrix} 1 & m \\ m & -1 \end{bmatrix}\begin{bmatrix} 1 & m \\ -m & 1 \end{bmatrix} \\ &= \frac{1}{1+m^2}\begin{bmatrix} 1-m^2 & 2m \\ 2m & m^2-1 \end{bmatrix}. \end{aligned}$$

Eigenvector & Eigenvalue

Definition. Let $T : V \to V$ be $\mathbb{F}$-linear. A nonzero vector $\mathbf{v} \in V \setminus \{\mathbf{0}\}$ is an **eigenvector** of T if $\exists \lambda \in \mathbb{F}$ such that

$$T(\mathbf{v}) = \lambda \mathbf{v} \in V.$$

The scalar λ is called the **eigenvalue** corresponding to $\mathbf{v}$.

Remark 1. If $\mathbf{v} \neq 0$ and $T(\mathbf{v}) = \lambda \mathbf{v}$, then the one-dimensional subspace $\mathbb{F}\mathbf{v}$ satisfying

$$\begin{array}{rccc} T|_{\mathbb{F}\mathbf{v}} & : & \mathbb{F}\mathbf{v} & \longrightarrow & \mathbb{F}\mathbf{v} \\ & & c\mathbf{v} & \longmapsto & \lambda(c\mathbf{v}) \end{array} \qquad (\because T(c\mathbf{v}) = cT(\mathbf{v}) = c\lambda\mathbf{v} = \lambda(c\mathbf{v})),$$

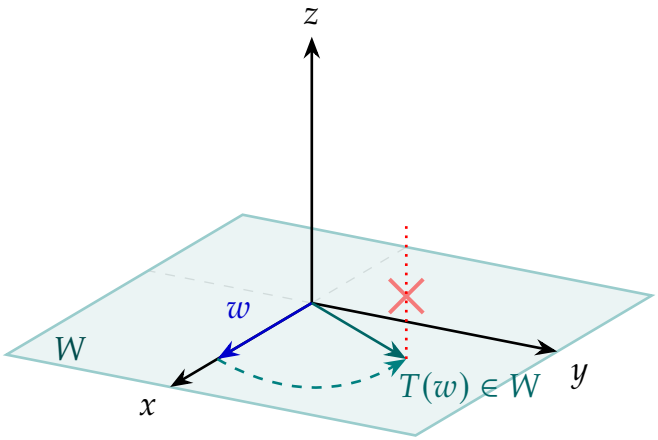
Equivalently, the restriction $T|_{\mathbb{F}\mathbf{v}} : \mathbb{F}\mathbf{v} \to \mathbb{F}\mathbf{v}$ acts as scalar multiplication by λ.

Remark 2 (T-invariant)**.** Let $T : V \to V$ be $\mathbb{F}$-linear. Let a subspace $W \leq V$ satisfy

$$T[W] \subseteq W \qquad (\iff \forall\, \mathbf{w} \in W,\ T(\mathbf{w}) \in W).$$

1. The restriction map
$$T|_W : W \to W, \qquad \mathbf{w} \mapsto T(\mathbf{w})$$
is a well-defined linear operator on W.

2. If $\dim V < \infty$ and $\mathcal{B} = (\mathbf{w}_1, \ldots, \mathbf{w}_k, \mathbf{v}_{k+1}, \ldots, \mathbf{v}_n)$ is a basis of V such that $(\mathbf{w}_1, \ldots, \mathbf{w}_k)$ is a basis of W, then
$$[T]_{\mathcal{B}} = \begin{pmatrix} A & * \\ 0 & B \end{pmatrix},$$
where A represents $T|_W$ and B represents the induced map on V/W.



$$\begin{array}{c} \\ x \\ y \\ z \end{array} \begin{array}{c} \begin{array}{cc} e_1 \in W & e_2 \in W \end{array} \\ \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{array}$$

Diagonalizability of Linear Operator

Definition 1. We say $T : V \to V$ is *diagonalizable* if $\exists$ a basis $\mathcal{B}$ of V such that $[T]_{\mathcal{B}}$ is diagonal.

Remark 3. T is diagonalizable if and only if V has a basis consisting of eigenvectors of T.
A diagonal matrix

$$\begin{bmatrix} d_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & d_n \end{bmatrix}$$

acts by scaling each basis vector $\mathbf{v}_i$ *independently*, and "scaling a nonzero vector" is the eigenvector condition $T(\mathbf{v}_i) = d_i\mathbf{v}_i$.

Note. Let $A \in \mathrm{Mat}_n(\mathbb{F})$. The associated linear map $L_A : \mathbb{F}^n \to \mathbb{F}^n$ is given by $L_A(\mathbf{x}) = A\mathbf{x}$.

Diagonalizability of Matrix

Definition 2. A matrix $A \in \mathrm{Mat}_n(\mathbb{F})$ is *diagonalizable over* $\mathbb{F}$ if $\exists P \in \mathrm{GL}_n(\mathbb{F})$ and a diagonal matrix D such that

$$D = P^{-1}AP.$$

Eigenbasis and Similarity

Proposition 1. *Let $A \in \mathrm{Mat}_n(\mathbb{F})$.*

(i) If $\mathcal{B} = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is a basis of $\mathbb{F}^n$ consisting of eigenvectors of A, and $P = [\mathbf{v}_1 \ \cdots \ \mathbf{v}_n]$, then $P^{-1}AP$ is diagonal with $P \in \mathrm{GL}_n(\mathbb{F})$.

(ii) Conversely, if $P^{-1}AP = D$ is diagonal, then the columns of P form an eigenbasis of A (with eigenvalues given by the diagonal entries of D).

Characteristic polynomial

Definition 3. For $A \in \mathrm{Mat}_n(\mathbb{F})$, the **characteristic polynomial** of A is

$$\chi_A(x) := \det(A - x\mathrm{Id}_n) \in \mathbb{F}[x].$$

Eigenvalues are roots

Proposition 2. *Let $A \in \mathrm{Mat}_n(\mathbb{F})$ and define the characteristic polynomial*

$$\chi_A(x) := \det(A - x\mathrm{Id}_n) \in \mathbb{F}[x].$$

A scalar $\lambda \in \mathbb{F}$ is an eigenvalue of A if and only if $\chi_A(x) = 0$.

Observation (Characteristic polynomial in 2×2). Let $A=\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathrm{Mat}_2(\mathbb{F})$. Then

$$\begin{aligned}\det(A-\lambda I_2) &= \det\begin{bmatrix} a-\lambda & b \\ c & d-\lambda \end{bmatrix} \\ &= (a-\lambda)(d-\lambda)-bc \\ &= ad-(a+d)\lambda+\lambda^2-bc \\ &= \lambda^2-(a+d)\lambda+(ad-bc) \\ &= \lambda^2-\mathrm{tr}(A)\lambda+\det(A).\end{aligned}$$

$$\chi_A(\lambda)=\lambda^2-(\mathrm{tr}A)\,\lambda+\det(A)$$

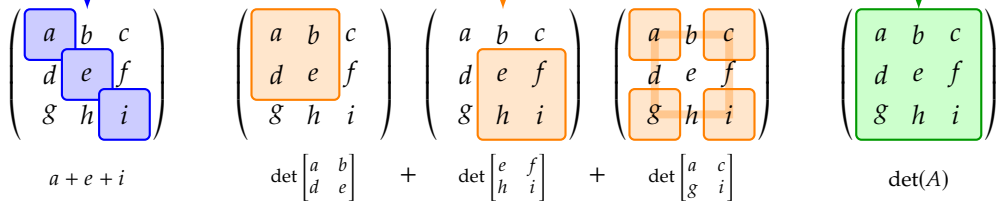


Observation (Characteristic polynomial in 3×3). Let $A=\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \in \mathrm{Mat}_3(\mathbb{F})$. Then

$$\det(A-\lambda I_3)=\det\begin{bmatrix} a-\lambda & b & c \\ d & e-\lambda & f \\ g & h & i-\lambda \end{bmatrix}=(a-\lambda)\det\begin{bmatrix} e-\lambda & f \\ h & i-\lambda \end{bmatrix}-b\det\begin{bmatrix} d & f \\ g & i-\lambda \end{bmatrix}+c\det\begin{bmatrix} d & e-\lambda \\ g & h \end{bmatrix}$$

$$\begin{aligned}&= (a-\lambda)\big((e-\lambda)(i-\lambda)-fh\big)-b\big(d(i-\lambda)-fg\big)+c\big(dh-ge+g\lambda\big) \\ &= (a-\lambda)\big(ei-(e+i)\lambda+\lambda^2-fh\big)-b\big(di-d\lambda-fg\big)+c\big(dh-ge+g\lambda\big) \\ &= \big(aei-a(e+i)\lambda+a\lambda^2-afh\big)-\big(ei\lambda-(e+i)\lambda^2+\lambda^3-fh\lambda\big) \\ &\quad -\big(bdi-bd\lambda-bfg\big)+\big(cdh-cge+cg\lambda\big) \\ &= -\lambda^3+(a+e+i)\lambda^2-(ae+ai+ei-fh-bd-cg)\lambda+(aei-afh-bdi-bfg+cdh-cge) \\ &= -\lambda^3+\mathrm{tr}(A)\lambda^2-\left(\det\begin{bmatrix} a & b \\ d & e \end{bmatrix}+\det\begin{bmatrix} a & c \\ g & i \end{bmatrix}+\det\begin{bmatrix} e & f \\ h & i \end{bmatrix}\right)\lambda+\det(A).\end{aligned}$$

$$\chi_A(\lambda)=-\lambda^3+c_2\lambda^2-c_1\lambda+c_0$$



Note (Principal minors). A principal minor of a square matrix is the determinant of a principal submatrix, which is formed by selecting the same set of indices for both rows and columns.

Let $A = (a_{ij}) \in \mathrm{Mat}_n(\mathbb{F})$ and $\{i_1 < i_2 < \cdots < i_\ell\} \subseteq \{1, \ldots, n\}$. For each $\ell = 1, \ldots, n$, we define

$$E_\ell(A) := \sum_{1 \le i_1 < \cdots < i_k \le n} \det \begin{bmatrix} a_{i_1,i_1} & a_{i_1,i_2} & \cdots & a_{i_1,i_k} \\ a_{i_2,i_1} & a_{i_2,i_2} & \cdots & a_{i_2,i_k} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i_k,i_1} & a_{i_k,i_2} & \cdots & a_{i_k,i_k} \end{bmatrix},$$

so

$$\begin{aligned} E_1(A) &:= \sum_{i=1}^{n} \det \begin{bmatrix} a_{ii} \end{bmatrix} = \sum_{i=1}^{n} a_{ii} = \mathrm{tr}(A), \\ E_2(A) &:= \sum_{1 \le i < j \le n} \det \begin{bmatrix} a_{ii} & a_{ij} \\ a_{ji} & a_{jj} \end{bmatrix}, \\ E_3(A) &:= \sum_{1 \le i < j < k \le n} \det \begin{bmatrix} a_{ii} & a_{ij} & a_{ik} \\ a_{ji} & a_{jj} & a_{jk} \\ a_{ki} & a_{kj} & a_{kk} \end{bmatrix}, \\ &\cdots, \\ E_n(A) &:= \det(A). \end{aligned}$$

Characteristic polynomial: Trace and Determinant as coefficients

Theorem 3. *Let $A = (a_{ij}) \in \mathrm{Mat}_n(\mathbb{F})$ and define*

$$\chi_A(x) := \det(A - x\mathrm{Id}_n) \in \mathbb{F}[x].$$

For $k = 1, \ldots, n$ define $E_k(A)$ by

$$E_k(A) := \sum_{1 \le i_1 < \cdots < i_k \le n} \det \left[a_{i_p i_q} \right]_{p,q=1}^{k},$$

so $E_1(A) = \sum_{i=1}^{n} a_{ii} = \mathrm{tr}(A)$ and $E_n(A) = \det(A)$. Then

$$\chi_A(x) = (-1)^n \left(x^n - E_1(A)x^{n-1} + E_2(A)x^{n-2} - \cdots + (-1)^n E_n(A) \right).$$

Proof. □

Determinant and Trace of a Linear Operator

Let $T : V \to V$ be a linear operator on a finite-dimensional vector space V over $\mathbb{F}$.

Determinant and trace of a linear operator

Definition 4. Choose any basis $\mathcal{B}$ of V. Define

$$\det(T) := \det([T]_{\mathcal{B}}), \qquad \operatorname{tr}(T) := \operatorname{tr}([T]_{\mathcal{B}}).$$

Remark 4. We must check that these definitions do not depend on the choice of basis.

Similarity preserves Determinant and Trace

Lemma 4. *Let $A, B \in \mathrm{Mat}_{n\times n}(\mathbb{F})$. Suppose there exists an invertible matrix P such that*

$$A = P^{-1}BP.$$

Then

$$\det(A) = \det(B), \qquad \operatorname{tr}(A) = \operatorname{tr}(B).$$

Proof. For the determinant,

$$\det(A) = \det(P^{-1}BP) = \det(P^{-1})\det(B)\det(P) = \det(B).$$

For the trace,

$$\operatorname{tr}(A) = \operatorname{tr}(P^{-1}BP) = \operatorname{tr}(BPP^{-1}) = \operatorname{tr}(B).$$

since $\operatorname{tr}(XY) = \operatorname{tr}(YX)$ for square matrices X, Y. □

Remark 5. If $\mathcal{B}$ and $\mathcal{B}'$ are two bases of V, then $[T]_{\mathcal{B}}$ and $[T]_{\mathcal{B}'}$ are similar matrices. Therefore determinant and trace are well-defined invariants of the operator T.

Eigenvalues and the Determinant Condition

Recall that $\mathbf{u} \in V$ is an eigenvector of T with eigenvalue λ if and only if

$$\mathbf{u} \neq 0, \qquad (T - \lambda \mathrm{id}_V)(\mathbf{u}) = 0.$$

Thus

$$\lambda \text{ is an eigenvalue} \iff \ker(T - \lambda \mathrm{id}_V) \neq \{\mathbf{0}\}.$$

Proposition 5. *Let $T : V \to V$ be a linear operator on a finite-dimensional vector space. Then*

$$\lambda \in \mathbb{F} \text{ is an eigenvalue of } T \iff \det(T - \lambda \mathrm{id}_V) = 0.$$

Proof. Choose a basis $\mathcal{B}$ of V. Then

$$[T - \lambda \mathrm{id}_V]_\beta = [T]_\beta - \lambda I.$$

Now

$$\ker(T - \lambda \mathrm{id}_V) \neq \{0\}$$

if and only if the matrix $[T]_\beta - \lambda I$ has a nontrivial kernel. Since V is finite-dimensional, this happens if and only if the matrix is not invertible, which happens if and only if its determinant is zero. Hence

$$\lambda \text{ is an eigenvalue} \iff \det([T]_\beta - \lambda I) = 0.$$

But this determinant is exactly $\det(T - \lambda \mathrm{id}_V)$. □

Characteristic polynomial of a linear operator

Definition 5. The characteristic polynomial of T is

$$\varphi_T(t) = \det(T - t\mathrm{id}_V).$$

Remark 6. λ is an eigenvalue of $T \iff \varphi_T(\lambda) = 0$.

0.1 Linear Independence of Eigenvectors with Distinct Eigenvalues

Lemma 6. *Let $u_1, \ldots, u_k$ be eigenvectors of T with eigenvalues*

$$\lambda_1, \ldots, \lambda_k.$$

Assume the eigenvalues are mutually distinct:

$$\lambda_i \neq \lambda_j \quad \text{whenever} \quad i \neq j.$$

Then

$$\{u_1, \ldots, u_k\}$$

is linearly independent.

Proof. Suppose not. Let h be the smallest positive integer such that

$$\{u_1, \ldots, u_h\}$$

is linearly dependent. Then $h \geq 2$, and $\{u_1, \ldots, u_{h-1}\}$ is linearly independent.

There exist scalars $a_1, \ldots, a_h$, not all zero, such that

$$a_1u_1 + a_2u_2 + \cdots + a_hu_h = 0. \tag{1}$$

By minimality of h, we must have $a_h \neq 0$.

Apply T to (1). Since $T(u_i) = \lambda_i u_i$, we get

$$a_1\lambda_1u_1 + a_2\lambda_2u_2 + \cdots + a_h\lambda_hu_h = 0. \tag{2}$$

Multiply (1) by λ_h:

$$a_1\lambda_hu_1 + a_2\lambda_hu_2 + \cdots + a_h\lambda_hu_h = 0. \tag{3}$$

Subtract (3) from (2):

$$a_1(\lambda_1 - \lambda_h)u_1 + \cdots + a_{h-1}(\lambda_{h-1} - \lambda_h)u_{h-1} = 0.$$

Since $u_1, \ldots, u_{h-1}$ are linearly independent, each coefficient is zero:

$$a_i(\lambda_i - \lambda_h) = 0, \qquad 1 \leq i \leq h-1.$$

Because the eigenvalues are distinct, $\lambda_i - \lambda_h \neq 0$, so

$$a_i = 0, \qquad 1 \leq i \leq h-1.$$

Then (1) becomes

$$a_hu_h = 0.$$

Since $u_h \neq 0$, we get $a_h = 0$, contradiction.

Therefore the eigenvectors are linearly independent. □

Corollary 7. *Let* $\dim V = n$. *If* $T : V \to V$ *has* n *distinct eigenvalues in* $\mathbb{F}$, *then* T *is diagonalizable over* $\mathbb{F}$.

Proof. Choose one eigenvector for each eigenvalue. By the lemma, these n eigenvectors are linearly independent. Since $\dim V = n$, they form a basis of V. Hence V has a basis consisting of eigenvectors, so T is diagonalizable. □

Remark 7. The converse is false. A linear operator may be diagonalizable even if it has fewer than n distinct eigenvalues. The identity map on an n-dimensional vector space has only one eigenvalue, 1, but it is diagonalizable.

Eigenspaces

Eigenspace

Definition 6. Let λ be an eigenvalue of $T : V \to V$. Define

$$E_\lambda = \{u \in V : T(u) = \lambda u\}.$$

This is called the *eigenspace* corresponding to λ.

Remark 8. Since

$$T(u) = \lambda u \quad \Longleftrightarrow \quad (T - \lambda \mathrm{id}_V)(u) = 0,$$

we have

$$E_\lambda = \ker(T - \lambda \mathrm{id}_V).$$

Therefore E_λ is a subspace of V.

Exercise 1. Show directly that $E_\lambda \leq V$.

Solution. Let $u, v \in E_\lambda$ and $c \in \mathbb{F}$. Then

$$T(u + v) = T(u) + T(v) = \lambda u + \lambda v = \lambda(u + v),$$

so $u + v \in E_\lambda$. Also

$$T(cu) = cT(u) = c\lambda u = \lambda(cu),$$

so $cu \in E_\lambda$. Thus E_λ is a subspace. □

Rank-nullity and Eigenspace Dimension

For a linear map $T : V \to W$, the first isomorphism theorem gives

$$V/\ker(T) \cong \mathrm{Im}(T).$$

Taking dimensions gives

$$\dim V - \dim \ker(T) = \dim \mathrm{Im}(T).$$

Therefore

$$\dim V = \text{nullity}(T) + \text{rank}(T).$$

This is the rank-nullity theorem.

Applying this to $T - \lambda \text{id}_V$, we get

$$\dim E_\lambda = \dim \ker(T - \lambda \text{id}_V) = \dim V - \text{rank}(T - \lambda \text{id}_V).$$

The number

$$\dim E_\lambda$$

is called the *geometric multiplicity* of the eigenvalue λ.

Example 2 (A diagonalizable 2×2 matrix)**.** Let

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Then

$$\varphi_A(t) = \det(A - tI) = \begin{vmatrix} 1-t & 1 \\ 1 & 1-t \end{vmatrix} = (1-t)^2 - 1 = t^2 - 2t = t(t-2).$$

So the eigenvalues are

$$\lambda = 0, \qquad \lambda = 2.$$

For $\lambda = 0$,

$$E_0 = \ker(A) = \text{span}\left\{ \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}.$$

For $\lambda = 2$,

$$E_2 = \ker(A - 2I) = \text{span}\left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}.$$

Thus A has two linearly independent eigenvectors in $\mathbb{R}^2$, so it is diagonalizable.

Example 3 (Diagonalizable with repeated eigenvalue)**.** Let

$$A = \begin{pmatrix} 4 & 0 & 1 \\ 2 & 3 & 2 \\ 1 & 0 & 4 \end{pmatrix}.$$

Then

$$\varphi_A(t) = \det(A - tI) = \begin{vmatrix} 4-t & 0 & 1 \\ 2 & 3-t & 2 \\ 1 & 0 & 4-t \end{vmatrix}.$$

Expanding along the second column,

$$\varphi_A(t) = (3-t)\begin{vmatrix} 4-t & 1 \\ 1 & 4-t \end{vmatrix} = (3-t)((4-t)^2-1).$$

Thus

$$\varphi_A(t) = (3-t)(t^2-8t+15) = (3-t)(t-3)(t-5) = -(t-3)^2(t-5).$$

The eigenvalues are

$$3, \qquad 5.$$

There are only two distinct eigenvalues, but A may still be diagonalizable.

For $\lambda = 5$,

$$A - 5I = \begin{pmatrix} -1 & 0 & 1 \\ 2 & -2 & 2 \\ 1 & 0 & -1 \end{pmatrix}.$$

The rank is 2, so

$$\dim E_5 = 3 - 2 = 1.$$

Solving gives

$$E_5 = \operatorname{span}\left\{\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}\right\}.$$

For $\lambda = 3$,

$$A - 3I = \begin{pmatrix} 1 & 0 & 1 \\ 2 & 0 & 2 \\ 1 & 0 & 1 \end{pmatrix}.$$

The rank is 1, so

$$\dim E_3 = 3 - 1 = 2.$$

Solving gives

$$E_3 = \operatorname{span}\left\{\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}\right\}.$$

Hence

$$\dim E_3 + \dim E_5 = 2 + 1 = 3 = \dim \mathbb{R}^3.$$

Therefore A is diagonalizable. One eigenbasis is

$$\beta = \left\{ \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \right\},$$

and with respect to this basis,

$$[L_A]_\beta = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{pmatrix}.$$

Sums and direct sums of subspaces

Sum of subspaces

Definition 7. Let $U, W \leq V$. Define

$$U + W = \{u + w : u \in U,\ w \in W\}.$$

This is called the sum of U and W.

Exercise 2. Show that

$$U + W \leq V$$

and

$$U \cap W \leq U + W.$$

Direct sum

Definition 8. Let $U, W \leq V$. We say

$$V = U \oplus W$$

if

$$V = U + W$$

and

$$U \cap W = \{0\}.$$

Remark 9. Equivalently,

$$V = U \oplus W$$

means that every vector $v \in V$ can be written uniquely as

$$v = u + w, \qquad u \in U, \quad w \in W.$$

More generally, for subspaces $W_1, \ldots, W_m \leq V$, we write

$$V = W_1 \oplus W_2 \oplus \cdots \oplus W_m$$

if every vector $v \in V$ can be written uniquely as

$$v = w_1 + w_2 + \cdots + w_m, \qquad w_i \in W_i.$$

Equivalently,

$$V = W_1 + \cdots + W_m$$

and

$$W_j \cap \Big(W_1 + \cdots + W_{j-1} + W_{j+1} + \cdots + W_m \Big) = \{0\}$$

for each j.

Exercise 3. Let $u_1, \ldots, u_n$ be a basis for V. Let

$$U = \operatorname{span}\{u_1, \ldots, u_k\}, \qquad W = \operatorname{span}\{u_{k+1}, \ldots, u_n\}.$$

Show that

$$V = U \oplus W.$$

Diagonalization criterion using eigenspaces

Let $T : V \to V$ be a linear operator on an n-dimensional vector space. Suppose the distinct eigenvalues of T are

$$\lambda_1, \ldots, \lambda_k, \qquad k \leq n.$$

For each i, let

$$E_{\lambda_i} = \ker(T - \lambda_i \mathrm{id}_V).$$

Diagonalization theorem

Theorem 8. *The linear operator T is diagonalizable if and only if*

$$V = E_{\lambda_1} \oplus E_{\lambda_2} \oplus \cdots \oplus E_{\lambda_k}.$$

Proof. First suppose T is diagonalizable. Then there exists a basis β of V consisting of eigenvectors.

Group this basis according to eigenvalue:

$$\beta = \{u_{11}, \dots, u_{1m_1}, u_{21}, \dots, u_{2m_2}, \dots, u_{k1}, \dots, u_{km_k}\},$$

where each u_{ij} has eigenvalue λ_i.

Let

$$W_i = \text{span}\{u_{i1}, \dots, u_{im_i}\}.$$

Since β is a basis,

$$V = W_1 \oplus \cdots \oplus W_k.$$

Clearly

$$W_i \subseteq E_{\lambda_i}.$$

We claim that actually $W_i = E_{\lambda_i}$.

Let $v \in E_{\lambda_i}$. Since $V = W_1 \oplus \cdots \oplus W_k$, write uniquely

$$v = w_1 + \cdots + w_k, \qquad w_j \in W_j.$$

Then

$$T(v) = \lambda_i v.$$

But also, since T acts as multiplication by λ_j on W_j,

$$T(v) = \lambda_1 w_1 + \cdots + \lambda_k w_k.$$

Thus

$$0 = T(v) - \lambda_i v = \sum_{j=1}^{k} (\lambda_j - \lambda_i) w_j.$$

By uniqueness in the direct sum, each component must be zero. Hence for $j \neq i$,

$$(\lambda_j - \lambda_i) w_j = 0.$$

Since $\lambda_j \neq \lambda_i$, we get $w_j = 0$ for all $j \neq i$. Therefore $v = w_i \in W_i$. Thus

$$E_{\lambda_i} = W_i.$$

Hence

$$V = E_{\lambda_1} \oplus \cdots \oplus E_{\lambda_k}.$$

Conversely, suppose

$$V = E_{\lambda_1} \oplus \cdots \oplus E_{\lambda_k}.$$

For each i, choose a basis β_i of E_{λ_i}. Then

$$\beta = \beta_1 \cup \cdots \cup \beta_k$$

is a basis of V because the sum is direct. Every vector in β_i is an eigenvector with eigenvalue λ_i. Therefore β is a basis of V consisting of eigenvectors. Hence T is diagonalizable. □

Dimension criterion

Corollary 9. *Let $T : V \to V$ have distinct eigenvalues $\lambda_1, \ldots, \lambda_k$ in $\mathbb{F}$. Then T is diagonalizable over $\mathbb{F}$ if and only if*

$$\dim E_{\lambda_1} + \cdots + \dim E_{\lambda_k} = \dim V.$$

Proof. Eigenspaces corresponding to distinct eigenvalues have zero intersection in the direct-sum sense, by the linear-independence lemma. Therefore

$$E_{\lambda_1} + \cdots + E_{\lambda_k}$$

is automatically a direct sum. Thus it equals V if and only if its dimension equals $\dim V$, namely if and only if

$$\sum_{i=1}^{k} \dim E_{\lambda_i} = \dim V.$$

□

Remark 10. If

$$V = E_{\lambda_1} \oplus \cdots \oplus E_{\lambda_k}$$

and $m_i = \dim E_{\lambda_i}$, then with respect to a basis obtained by joining bases of the eigenspaces,

$$[T]_\beta = \begin{pmatrix} \lambda_1 I_{m_1} & & & 0 \\ & \lambda_2 I_{m_2} & & \\ & & \ddots & \\ 0 & & & \lambda_k I_{m_k} \end{pmatrix}.$$

This is a diagonal matrix whose diagonal entries are the eigenvalues repeated according to the dimensions of their eigenspaces.

Summary

- An eigenvector is a nonzero vector u satisfying $T(u) = \lambda u$.
- A linear operator is diagonalizable if and only if it has a basis consisting of eigenvectors.
- For a matrix A, diagonalization means finding P invertible and D diagonal such that

$$D = P^{-1}AP.$$

- The eigenvalues of A are the roots of

$$\varphi_A(t) = \det(A - tI).$$

- Eigenvectors corresponding to distinct eigenvalues are linearly independent.
- If $\dim V = n$ and T has n distinct eigenvalues, then T is diagonalizable.
- The eigenspace corresponding to λ is

$$E_\lambda = \ker(T - \lambda \mathrm{id}_V).$$

- The operator T is diagonalizable if and only if

$$V = \bigoplus_\lambda E_\lambda,$$

  where the sum runs over the distinct eigenvalues of T.
- Equivalently,

$$T \text{ is diagonalizable} \iff \sum_\lambda \dim E_\lambda = \dim V,$$

  provided all eigenvalues needed for diagonalization lie in the field $\mathbb{F}$.
- Diagonalizability depends on the field. A real rotation matrix with angle not 0 or π is not diagonalizable over $\mathbb{R}$, but it is diagonalizable over $\mathbb{C}$.

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